$\sum_{i=1}^{n} D_i^2 = 90$ and $n = 7$.

Rank correlation coefficient is

$$\rho = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n\left(n^2 - 1\right)}$$

$$= 1 - \frac{6 \times 90}{7\left(7^2 - 1\right)} = 1 - \frac{540}{7 \times 48} = 1 - \frac{540}{336} = 1 - 1.6071 = -0.6071$$

Interpretation: There is a negative correlation between equity shares and preference share prices. There is a strong disagreement between equity shares and preference share prices.

4.5.1 Repeated ranks

When two or more items have equal values (i.e., a tie) it is difficult to give ranks to them. In such cases the items are given the average of the ranks they would have received. For example, if two individuals are placed in the 8th place, they are given the rank $\frac{8+9}{2} = 8.5$ each, which is common rank to be assigned and the next will be 10; and if three ranked equal at the 8th place, they are given the rank $\frac{8+9+10}{3} = 9$ which is the common rank to be assigned to each; and the next rank will be 11.

In this case, a different formula is used when there is more than one item having the same value.

$$\rho = 1 - 6\left[\frac{\sum D_i^2 + \frac{1}{12}\left(m_1^3 - m_1\right) + \frac{1}{12}\left(m_2^3 - m_2\right) + \ldots}{n\left(n^2 - 1\right)}\right]$$

where m_i is the number of repetitions of ith rank

Example 4.6

Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in the Commerce and Mathematics.

Marks in Commerce	15	20	28	12	40	60	20	80
Marks in Mathematics	40	30	50	30	20	10	30	60

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